

Vectored Thrust of a Ducted Fan with Jet Vanes by a Two-Way Coupled Rotating-Frame Vortex Lattice Method

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Control of a ducted-fan VTOL aircraft at hover rests on vanes immersed in the propulsor's own jet: at zero airspeed the propwash is the only dynamic pressure available to them. This paper presents a potential-flow method that resolves the complete propulsive wrench of such a system — rotor, duct, and exit vanes — in seconds per operating point. The rotor is solved as a vortex lattice in the blade-fixed rotating frame, with blades meshed on their CST camber surfaces wrapped on radius cylinders and trailing legs prescribed along the local relative wind, which keeps the hover limit well posed. The duct is paneled into the same solve as constant-strength source rings, a two-way interaction converged with the blade circulation. The exit vanes cannot share the rotating frame; they are solved in the inertial frame, reading the rotor through a time-mean slipstream reconstructed from the converged radial loading by annular momentum theory, with a viscous sectional coupling that handles their separation-dominated hover aerodynamics. A block Gauss–Seidel alternation closes the loop between the two frames, constrained by a swirl momentum budget: the vanes cannot extract more torque than the jet's angular-momentum flux — the shaft torque — delivers. On a reference cruciform configuration at hover the budget binds, the vanes recovering 99% of the available angular momentum where an uncoupled scheme reported three times that, and the converged control response about neutral is nearly antisymmetric. The method is implemented in the open-source C++23 toolkit Aeolion.

Nomenclature

c	section chord
c_l	section lift coefficient
M_x	rolling moment about the rotation axis
p	body-axis roll rate
Q	shaft torque
q	dynamic pressure
r	radial station
R	rotor tip radius
s	swirl budget factor
T	thrust
v_i	annulus axial induced velocity
V_{ax}	axial onset velocity at the disk
V_{kin}	kinematic (rotational + freestream) local velocity
V_{rel}	local relative velocity
w_i	annulus swirl velocity
α_{eff}	effective section angle of attack
Γ	bound circulation
λ	inflow ratio
σ	source panel strength
Ω	rotor angular speed

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I. Introduction

Tail-sitter VTOL aircraft of the ducted-fan class obtain control at hover from vanes placed in the duct's exit jet. Their sizing question is awkward for the established toolchain: at zero airspeed a conventional vortex lattice or panel method has no freestream dynamic pressure to put on a control surface, so vane authority, the thrust cost of vectoring, and the recovered swirl torque are outside such tools' model entirely, while RANS CFD of the full rotor-duct-vane system is too slow for the configuration loop. Classical VLM codes such as AVL¹ treat the airframe with, at most, a prescribed actuator-disk propulsor; isolated-rotor methods such as XROTOR² carry no airframe and no vanes.

This paper describes a method that closes that gap at potential-flow-plus-sectional-viscous fidelity, resolving the coupled rotor-duct-vane system in seconds. Three ingredients are presented in turn: the rotating-frame vortex lattice for the rotor (Sec. II), the two-way ducted solve and the inertial-frame vane solve with its reconstructed slipstream (Secs. III, IV), and the two-way rotor-vane alternation with its swirl momentum budget (Sec. V). Results on a reference cruciform configuration at hover follow in Sec. VI. The method is implemented in Aeolion, an open-source C++23 aerodynamics toolkit; the mathematical elements — horse-shoe vortices and constant-strength source panels — are the textbook ones,³ and the contribution is the coupling.

II. Rotor: Vortex Lattice in the Rotating Frame

The rotor is the same vortex lattice as a wing, posed in the blade-fixed rotating frame. Rotation is not special-cased in the solver: it enters as the body roll rate $p = \Omega$, whose kinematic velocity $\mathbf{V}_{kin} = \mathbf{V}_\infty - \boldsymbol{\omega} \times \mathbf{r}$ hands every blade panel its true Ωr tangential onset flow. The near-field Kutta-Joukowski force evaluation uses that same local velocity, so the body-axis load sums are dimensional propeller loads directly: thrust $T = -F_x$, shaft torque $Q = -M_x$, power $P = Q \Omega$.

Each blade is meshed with one Weissinger row per radial strip on the blade's CST camber surface,⁴ with two geometric choices that proved load-bearing rather than cosmetic:

1. *Chords wrap on their radius cylinders* — the surface a blade element actually sweeps — rather than lying in a flat tangent plane. Near a small hub, where chord is comparable to radius, a tangent-plane chord subtends tens of degrees of azimuth, and the phantom incidence it picks up can overwhelm the real camber and twist loading.
2. *Trailing legs leave along the local relative wind* at their roots — the tangential sweep plus the axial inflow, floored at the momentum-theory hover inflow $\lambda \approx 0.07$ — a prescribed linearized helix. At hover the local wind is almost purely tangential: trailing the wake axially instead leaves the legs near-perpendicular to the flow and the circulation solve diverges under refinement, while without the inflow floor a straight tangential leg lies in the rotor plane forever, crossing every radius outboard of its root.

The bare lattice is quasi-steady with a prescribed straight-line wake (no roll-up or contraction), carries induced torque only, and has no stall. The last two limitations are removed by a sectional viscous-inviscid coupling: per radial strip, the lattice provides α_{eff} from the local kinematic-plus-induced velocity, a 2-D section model provides $c_l(\alpha_{eff}, Re, Ma)$, the target circulation follows from the Kutta-Joukowski relation $|\Gamma| = \frac{1}{2} V_{rel} c c_l$, and the fixed point

$$R_i(\Gamma) = c_{l,i}^{VLM}(\Gamma) - c_{l,i}^{sect}(\alpha_{eff,i}(\Gamma), Re_i, Ma_i) = 0 \quad (1)$$

is converged with type-II Anderson acceleration⁵ (67 plain-relaxation iterations reduced to 9 on the reference rotor), guarded by a kinematic-speed circulation cap that breaks the $\Gamma \rightarrow V_{rel} \rightarrow \Gamma$ runaway. The section model is an interface seam: an analytic viscous polar, or a transpiration-coupled integral boundary-layer solver (Thwaites,⁶ Head,⁷ Squire-Young⁸) whose displacement effect enters the 2-D section problem as transpiration boundary conditions. The coupling caps the stalled hover root loading the inviscid lattice overpredicts and puts profile torque into the shaft-power sum.

III. The Ducted Rotor

The duct is meshed as a closed annular ring of constant-strength source panels (outer wall, bore wall, two caps) centered on the rotor plane, and passed into the coupled solve as its source system. The interaction is two-way: every coupling iteration re-solves the source strengths against the current blade circulation (the duct’s no-through-flow condition sees the propwash; the source-only influence block is factored once), and the strip velocities include the duct’s induced flow (the blades see the bore’s constriction). One frame subtlety makes this legitimate: in the blade-fixed frame the duct rotates backwards, but a body of revolution about the rotation axis moves only tangentially to itself, so its boundary condition is unchanged (exactly, up to sector faceting).

The duct’s own load is the surface-pressure integral $\mathbf{F} = \sum (\frac{1}{2}\rho|V|^2 - q_\infty)A\hat{\mathbf{n}}$, well behaved at hover where the gauge q_∞ is negligible — and this is where a shroud earns its keep: the bore-lip suction ahead of the disk points upstream.

IV. Exit Vanes and the Reconstructed Slipstream

The exit vanes are all-moving flat plates at the duct exit, spanning radially on their hinge axes and deflecting as rigid rotations about them. They cannot share the rotor’s frame: a non-axisymmetric surface swept backwards through a rotating frame has a time-dependent boundary condition. The solve is therefore partitioned by frame — the vanes are meshed as ordinary lifting surfaces and solved in the inertial frame ($p = 0$), reading the rotor system through a prescribed external slipstream.

That slipstream is *reconstructed*, not field-averaged. The quasi-steady rotor’s prescribed straight trailing legs never wrap into the downstream helical wake cylinder, so their Biot–Savart mean carries swirl but almost no axial jet a few chords aft of the disk; vanes placed there would see no dynamic pressure. Instead, annular momentum theory is applied to the loads the lattice actually converged: per annulus,

$$dT = 4\pi r \rho (V_{ax} + v_i) v_i dr \Rightarrow v_i, \quad dQ = 4\pi r^3 \rho (V_{ax} + v_i) w_i dr \Rightarrow w_i, \quad (2)$$

with dT , dQ summed over blades from the per-strip forces, the axial component developing toward the classical far-wake doubling over about a diameter, the swirl carried unchanged, and the duct’s exact source induction added on top.

The vanes receive the same viscous sectional coupling as the blades, with the boundary-layer section model by default and each trailing leg leaving along the *local mean flow* at its root — in the jet, helically, with the swirl. This matters twice over. First, at hover the swirl alone puts the vanes near 16–20° of local incidence before any command, so their hover aerodynamics are separation-dominated by nature; the section solver resolves attached and mildly separated flow, blends into a saturated polar, and applies a flat-plate deep-stall decay beyond it. Second, two numerical safeguards proved necessary: the coupling residual is weighted by local dynamic pressure (a vane tip parked at 85° incidence in the jet’s shear edge at 7% of the jet’s q must not hold the solve hostage), and the deep-stall decay is what makes such strips satisfiable at all.

The propulsive wrench is the sum of the two solves’ near-field loads about the hub: rotor circulatory and profile forces, duct surface pressure, vane circulatory and profile forces. The vanes’ drag debits the net thrust; their side forces and M_y/M_z are the control authority — real at zero airspeed, the propwash being their only dynamic pressure — and even undeflected they recover part of the swirl as counter-torque.

V. Two-Way Coupling and the Swirl Momentum Budget

Partitioned as above, the scheme is one-way: the vanes read the jet, the rotor never feels the vanes, and nothing depletes the prescribed swirl — a high-solidity vane set can recover more torque than the jet carries. The two-way scheme closes the loop as a block Gauss–Seidel alternation between the frames. Per outer pass:

1. *Rotor + duct solve* (rotating frame), with the vanes entering through the external-velocity hook as the azimuthal mean of their induced field,

$$\bar{\mathbf{v}}_{vane}(P) = \frac{1}{K} \sum_{j=0}^{K-1} R_x(-\theta_j) \mathbf{v}_{vane}(R_x(\theta_j)P), \quad \theta_j = 2\pi j/K. \quad (3)$$

A blade sweeps past the static vane system once per revolution, and its quasi-steady condition sees the time mean of that encounter. Averaging is legitimate in *this* direction because the vane system's near field is fully represented by its explicit legs — it is exactly the operation that fails in the other direction.

2. *Slipstream reconstruction* from the fresh rotor loading, Eq. (2), with the swirl scaled by the budget factor $s \in (0, 1]$.
3. *Vane remesh and solve* (inertial frame) — remesh, because the trailing legs follow the local mean flow, which just changed.
4. *The swirl budget*. The angular-momentum flux the jet delivers is the shaft torque, $\dot{L}_{jet} = Q = -M_x^{rotor}$; the vanes cannot remove more than arrives. Whenever the extracted M_x^{vane} exceeds it, the swirl factor contracts,

$$s \leftarrow s \cdot \frac{Q}{M_x^{vane}}, \quad (4)$$

whose fixed point is the budget-saturated state $M_x^{vane} = Q$ — vanes that de-swirl the jet essentially completely.

5. *Relaxation and convergence*. The vane circulation fed back to the rotor is under-relaxed; the outer residual is the relative change of the total wrench together with the vane-circulation change.

Still outside the model, by construction: unsteady blade-passing loads (every exchange is a time mean), the vanes' straight-line wakes, and radially resolved swirl depletion — the budget is a single scalar, the natural refinement being a band-by-band bookkeeping.

VI. Results

Results are shown for a reference two-blade rotor with a snug shroud (4% tip clearance) and a four-vane cruciform at the duct exit, at static (hover) conditions.

DUCTED ROTOR. At hover the shroud lifts thrust from 7.60 N open to 8.82 N ducted (+16%), the duct itself carrying +1.17 N of bore-lip suction, with the coupled solve converged in 9 iterations; the static comparison across rpm (Fig. 1) shows the ~15% augmentation at essentially unchanged shaft torque. A loosely fitted ring with large tip clearance correctly earns almost nothing — the augmentation is the interaction, not the presence of a ring.

VANE SYSTEM AND THE BUDGET. On the reference cruciform at hover the outer alternation converges in 7 passes. The swirl budget binds hard: the swirl factor converges near $s = 0.33$, and the vanes recover 99% of the jet's angular-momentum flux — the budget-saturated, essentially complete de-swirl state — where the one-way scheme had reported three times the available torque. The rotor measurably feels the vanes' blockage (a thrust shift, a perturbation rather than an upheaval), and with the budgeted swirl bias small, the control response about neutral is nearly antisymmetric: +0.100 / -0.104 N of side force at $\pm 8^\circ$ of vane deflection. Wake direction matters at this level: pointing the vane wakes along the local mean flow (helically, with the swirl) rather than hardcoded downstream roughly doubled the control response about the swirl-biased neutral and strengthened the undeflected counter-torque.

VERIFICATION. Every claim above is pinned by the automated test suite: exact cruciform symmetry undeflected, counter-torque opposing the rotor's, the budget bound $M_x^{vane} \leq Q$ at convergence, opposite commands pulling the wrench opposite ways, drag cost for commanding against the swirl, and the survival of the control response under the two-way coupling.

VII. Conclusion

A ducted fan with exit vanes can be resolved as one coupled potential-flow system — rotating-frame vortex lattice for the blades, source rings for the duct, inertial-frame lifting surfaces for the vanes reading a momentum-theory slipstream reconstructed from the solved loading — with a scalar angular-momentum

Static thrust — ducted vs open propeller

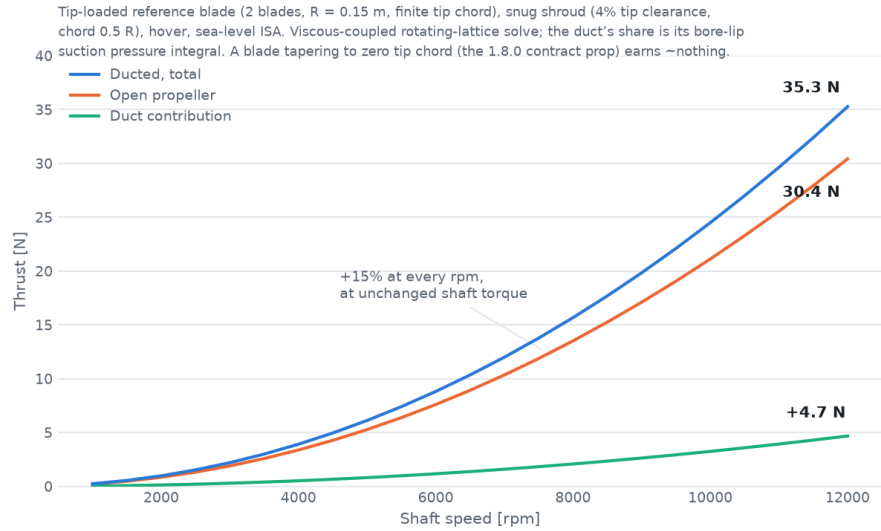


Figure 1. Static thrust versus rpm, ducted versus open rotor, with the duct's own contribution shown separately.



Figure 2. The reference configuration: aft view down the bore showing the four-vane cruciform around the hub (left), and side view with the vane chords running downstream from their exit-plane hinge lines (right); blades and vanes colored by solved circulation.

budget closing the swirl exchange between the frames. The budget is not a refinement but a correction of leading order: the unconstrained scheme overpredicted the recovered vane torque by a factor of three on the reference configuration, while the coupled scheme converges to the physically mandated budget-saturated state in a handful of outer passes and seconds of wall time. The method gives a ducted-fan VTOL designer the quantities the configuration loop actually needs at hover — net thrust with the vanes' drag debit, control moments about the swirl-biased neutral, and the recovered counter-torque — from a parametric geometry description, at interactive speed.

Acknowledgments

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